Energy of $\vec{E}$-field

$$\mathcal{E} = \frac{1}{2}\varepsilon_0 \int dV\, \vec{E}^2$$

Note : $\vec{E}^2 \sim \frac{1}{r^4}$ for point charge $\Rightarrow \int \frac{dr \cdot r^2}{r^4} \sim \int \frac{dr}{r^2}$ very strong divergence $\Rightarrow$ this is infinite energy of self-interaction of the charge with itself.

Solution: QM invalides classical at much greater scales

$$\frac{1}{4\pi\varepsilon_0}\frac{e^2}{a} \sim mc^2 \Rightarrow a \sim \frac{e^2}{4\pi\varepsilon_0 mc^2} = \underbrace{\left(\frac{e^2}{4\pi\varepsilon_0 \hbar c}\right)^2}_{\alpha^2 = \frac{1}{(137)^2}} \underbrace{\frac{4\pi\varepsilon_0 \hbar^2}{me^2}}_{a_B \sim 0.5\,\text{Å}}$$

$\Rightarrow a \ll a_B$ But quantum effects are important already on Bohr scale $\Rightarrow$ no real inconsistency!

$$\mathcal{E} = -\frac{1}{2}\varepsilon_0 \int dV\, \vec{\mathcal{E}} \cdot \vec{\nabla}\varphi = -\frac{1}{2}\varepsilon_0 \underbrace{\int dV\, \vec{\nabla}\cdot(\vec{E}\varphi)}_{\text{surface integral}} + \frac{1}{2}\varepsilon_0 \int dV\, \varphi \underbrace{\vec{\nabla}\cdot\vec{E}}_{\frac{\rho}{\varepsilon_0}}$$

$$= \frac{1}{2}\int dV\, \rho(r)\,\varphi(r)$$

$$= \frac{1}{2}\sum_n{}' q_n\, \varphi(\vec{r}_n) \quad \leftarrow \text{without self-interaction}$$

$$= \frac{1}{8\pi\varepsilon_0}\sum_{m\neq n} \frac{q_m q_n}{|\vec{r}_m - \vec{r}_n|} \quad \left(\text{every } n,m \text{ pair appears twice}\right)$$

Multipole expansion

Let charges occupy small region of size a
while the field it produces is observed at $R \gg a$.
Then can develop expansion in powers $\frac{a}{R}$.
Especially useful if total charge of the system is
$Q = \sum_n q_n = 0$. the integral is over volume where $\rho(r) \neq 0$

$$\varphi(\vec{R}) = \frac{1}{4\pi\varepsilon_0} \int dv\, \rho(\vec{r}) \frac{1}{|\vec{R}-\vec{r}|}$$

$$\frac{1}{|\vec{R}-\vec{r}|} = \frac{1}{R} - r_i \frac{\partial}{\partial R_i} \frac{1}{R} + \frac{1}{2} r_i r_j \frac{\partial^2}{\partial R_j \partial R_i} \frac{1}{R} + \dots$$

more convenient

Since $\frac{1}{|\vec{R}-\vec{r}|} \cong \frac{1}{R} + r_i \frac{\partial}{\partial r_i} \frac{1}{R} + \dots$

but $\frac{\partial}{\partial r_i} = - \frac{\partial}{\partial R_i}$

$$\frac{\partial}{\partial R_i} R = -\frac{1}{R^2} \frac{\partial R}{\partial R_i}, \quad \frac{\partial R}{\partial R_i} = \frac{\partial}{\partial R_i} \sqrt{\sum_j R_j^2} = \frac{R_i}{R}$$

$$\frac{\partial R_j}{\partial R_i} = \delta_{ij}$$

$$\Rightarrow \frac{\partial}{\partial R_i} \frac{1}{R} = -\frac{R_i}{R^3}$$

$$\frac{\partial}{\partial R_j} \frac{R_i}{R^3} = \frac{\delta_{ij}}{R^3} - 3 \frac{R_i}{R^4} \frac{\partial R}{\partial R_j} = \frac{R^2 \delta_{ij} - 3 R_i R_j}{R^5}$$

$$\Rightarrow \frac{1}{|\vec{R}-\vec{r}|} = \underbrace{\frac{1}{R}}_{\text{monopole}} + \underbrace{\frac{r_i R_i}{R^3}}_{\text{dipole}} + \underbrace{\frac{1}{2} r_i r_j \frac{3 R_i R_j - \delta_{ij} R^2}{R^5}}_{\text{quadrupole}} + \underbrace{\dots}_{\text{octupole}}$$

$$\Rightarrow \varphi(\vec{R}) = \frac{1}{4\pi\varepsilon_0} \left\{ \frac{1}{R} \underbrace{\int dv\, \rho(\vec{r})}_{Q} + \frac{\vec{R}}{R^3} \cdot \underbrace{\int dv\, \vec{r}\, \rho(\vec{r})}_{\vec{d} = \text{dipole moment}} + \dots \right\}$$

Dipole contribution is important if $Q=0$

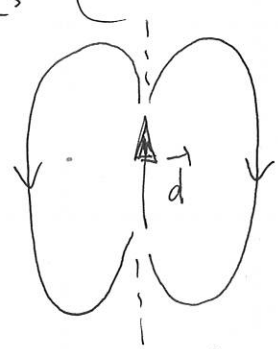
$$\varphi_{dip}(\vec{R}) = \frac{1}{4\pi\varepsilon_0}\frac{\vec{R}\cdot\vec{d}}{R^3}$$

$$\vec{E}_{dip}(\vec{R}) = -\vec{\nabla}\varphi_{dip} = -\frac{1}{4\pi\varepsilon_0}\hat{x}_i\frac{\partial}{\partial R_i}\frac{R_j d_j}{R^3} =$$

$$= -\frac{1}{4\pi\varepsilon_0}\hat{x}_i\left(\frac{\delta_{ij}\, d_j}{R^3} - \frac{3R_j d_j}{R^4}\frac{R_i}{R}\right) \qquad \begin{array}{l} R_i\hat{x}_i = \vec{R} \\ d_i\hat{x}_i = \vec{d} \end{array}$$

$$= +\frac{1}{4\pi\varepsilon_0}\frac{3\vec{R}(\vec{R}\cdot\vec{d}) - R^2\vec{d}}{R^5}$$

$$E_{dip} \sim \frac{2R^2}{R^5}\left(\text{for } \vec{R}\parallel\vec{d}\right), \sim \frac{-R^2}{R^5} \text{ for } \vec{R}\perp\vec{d}$$



$$\vec{d} = \int dV\, \rho(\vec{r})\,\vec{r} = \int dV\, \vec{\rho}(\vec{r})\left(\vec{r}+\vec{r}_0\right) = \vec{d} \text{ since } \vec{r}_0\int dV\rho = 0.$$

Monopole

Dipole

$$\vec{d} = +q\,\vec{r}_1 - q\,\vec{r}_2 = q\left(\vec{r}_1 - \vec{r}_2\right)$$

from $(-q)$ to $(+q)$

Quadrupole

$$\varphi_Q(\vec{R}) = \frac{1}{4\pi\varepsilon_0} \frac{3R_iR_j - \delta_{ij}R^2}{2R^5} \int dV\, \rho(\vec{r}) \Big(r_i r_j - \underbrace{\tfrac{1}{3}\delta_{ij} r^2}_{\text{added by hand}}\Big)$$

sum over repeated indices

Tensor $\quad 3R_iR_j - \delta_{ij}R^2 \rightarrow \begin{pmatrix} 3R_1^2 - R^2 & 0 & 0 \\ 0 & 3R_2^2 - R^2 & 0 \\ 0 & 0 & 3R_3^2 - R^2 \end{pmatrix}$

$$\text{Trace}(\quad) = 3(R_1^2 + R_2^2 + R_3^2) - 3R^2 = 0.$$

$$\sum_{ij} \delta_{ij} (3R_iR_j - \delta_{ij}R^2) = 3\sum_i R_i^2 - \sum_i \delta_{ii} R^2 = 0$$

Next, $\quad \left(r_i r_j - \frac{1}{3}\delta_{ij} r^2\right)\left(3R_iR_j - \delta_{ij}R^2\right) = \left(3 r_i r_j - \delta_{ij} r^2\right) R_i R_j$

$$- R^2 \underbrace{\delta_{ij}\left(r_i r_j - \tfrac{1}{3}\delta_{ij} r^2\right)}_{\phi}$$

$$\Rightarrow \varphi_Q(\vec{R}) = \frac{1}{4\pi\varepsilon_0} \frac{R_iR_j}{R^5} Q_{ij}$$

quadrupole moment $\quad Q_{ij} = \frac{1}{2}\int dV\, \rho(\vec{r}) \left(3 r_i r_j - \delta_{ij} r^2\right)$

$\overbrace{Q_{ji} = Q_{ij}}^{\text{symmetric}}, \quad \text{Tr}\,\hat{Q} = \sum_i Q_{ii} = 0$

$-q \quad 0 \quad +q$

$+q \quad 0 \quad -q$

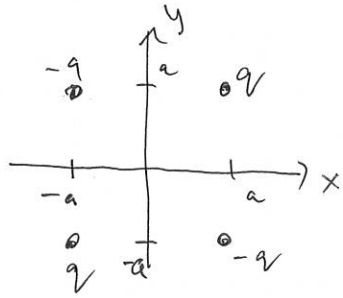
$$\Rightarrow \hat{Q} = \begin{pmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & \underline{Q_{22}} & \underline{Q_{23}} \\ Q_{31} & Q_{32} & Q_{33} \end{pmatrix}$$

Independent components are <u>underlined</u>

$Q_{33} = -Q_{11} - Q_{22}$

$$\vec{E}_Q(R) = -\frac{1}{4\pi\varepsilon_0} \hat{x}_k \frac{\partial}{\partial R_k} \frac{R_iR_jQ_{ij}}{R^5} = -\frac{1}{4\pi\varepsilon_0} \hat{x}_k \left(\frac{\delta_{ik}R_j + \delta_{jk}R_i}{R^5} - 5\frac{R_iR_j}{R^6}\frac{R_k}{R}\right) Q_{ij} = \frac{1}{4\pi\varepsilon_0} \frac{5Q_{ij}R_iR_jR_k - 2Q_{kj}R_jR^2}{R^7} \hat{x}_k$$

Example



$$Q_{ij} = \frac{1}{2} \sum_n q_n \left(3 r_i^{(n)} r_j^{(n)} - \delta_{ij} r_n^2\right)$$

charge index

$$Q_{xx} = \frac{1}{2}\Big\{ q\,(3a^2 - \underbrace{2a^2}_{\vec{r}_1^{\,2}}) + q\,(3(-a)^2 - 2a^2)$$

$$- q\,(3a^2 - 2a^2) - q\,(3(-a)^2 - 2a^2)\Big\} = 0 = Q_{yy} \Rightarrow Q_{zz} = 0$$

$$Q_{xy} = \frac{1}{2}\Big\{ q\,3a^2 + q\,3(-a)(-a) - q\,3(-a)(a) - q\,3(a)(-a)\Big\}$$

$$= \frac{1}{2}\, q \cdot 12a^2 = 3qa^2$$

$$Q_{xz} = Q_{yz} = 0$$

$$\Rightarrow \hat{Q} = 3qa^2 \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

It's electric field is as follows:

$$E_Q^x = \frac{1}{4\pi\varepsilon_0} \frac{5\left(Q_{xy} R_x R_y + Q_{yx} R_y R_x\right) R_x - 2 Q_{xy} R_y R^2}{R^7}$$

$$= \frac{1}{4\pi\varepsilon_0} \frac{2 Q_{xy} R_y \left(5R_x^2 - R^2\right)}{R^7}$$

$$E_Q^y = \frac{1}{4\pi\varepsilon_0} \frac{10 Q_{yx} R_y R_x R_y - 2 Q_{yx} R_x R^2}{R^7}$$

$$= \frac{1}{4\pi\varepsilon_0} 2 Q_{xy} R_x \frac{\left(5R_y^2 - R^2\right)}{R^7}$$

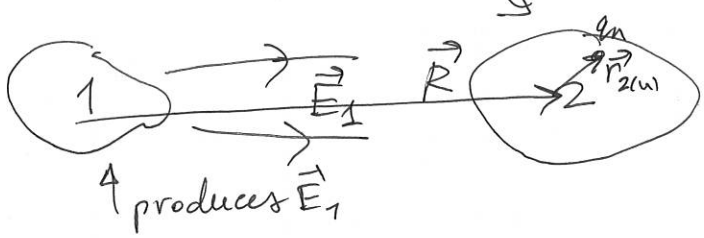
zero on high symmetry lines such as ones with $R_x = 0, R_y = 0$

$$E_Q^z = \frac{1}{4\pi\varepsilon_0} 10 Q_{xy} \frac{R_x R_y R_z}{R^7}$$

Interaction energy of multipole moments

Consider now multipole system in external potential

(produced by other system of charges, for example)



Interaction energy of system 2 with system 1

$$U_{12} = \sum q_2 \varphi_1$$

sum over charges in 2

$$U_{12} = \sum_n q_{2(n)} \varphi_1(\vec{R} + \vec{r}_{2(n)}) = \sum_n q_{2(n)} \Big\{ \varphi_1(\vec{R}) + \vec{r}_{2(n)} \cdot \frac{\partial}{\partial \vec{R}} \varphi_1(\vec{R}) + \frac{1}{2} r_{2(n)i}\, r_{2(n)j} \frac{\partial^2}{\partial R_i \partial R_j} \varphi_1(\vec{R}) + \dots \Big\}$$

$$= \varphi_1(\vec{R}) \sum_n q_{2(n)} - \vec{E}_1(\vec{R}) \cdot \underbrace{\sum_n q_{2(n)} \vec{r}_{2(n)}}_{\vec{d}_2}$$

$$+ \frac{1}{2} \frac{\partial^2 \varphi_1(\vec{R})}{\partial R_i \partial R_j} \sum_n q_{2(n)} \Big(r_{2(n)i}\, r_{2(n)j} - \underbrace{\tfrac{1}{3} \delta_{ij} \vec{r}_{2n}^{\,2}}_{\text{added by hand}} \Big)$$

Observe that $\dfrac{\partial^2 \varphi_1}{\partial R_i \partial R_j} \delta_{ij} = \nabla_R^2 \varphi_1 = -\underbrace{\vec{\nabla}_R \cdot \vec{E}_1(\vec{R})}_{=0}$

because there are no charges from (1) at this location!

$\Rightarrow$ therefore $\frac{1}{3} \underbrace{\frac{\partial^2 \varphi_1(\vec{R})}{\partial R_i \partial R_j}}_{-\frac{\partial E_{1i}}{\partial R_j}} \underbrace{\frac{1}{2} \sum_n \left(q_{2(n)} 3 r_{2(n)j} r_{2(n)i} - \delta_{ij} \vec{r}^{\,2}_{2(n)} \right)}_{Q_{ij}^{2(n)}}$

Interaction with quadrupole moment is like that:

$$-\frac{1}{3} \frac{\partial E_{1i}(\vec{R})}{\partial R_j} Q_{ij}$$

It is determined by the <u>gradient</u> of the external field at the location of the quadrupole!

Dipole-dipole interaction:

$$U = -\vec{E}_1 \cdot \vec{d}_2$$

$$\hookrightarrow \frac{1}{4\pi\varepsilon_0} \frac{3\vec{R}(\vec{R}\cdot\vec{d}_1) - R^2 \vec{d}_1}{R^5}$$

$$U = +\frac{1}{4\pi\varepsilon_0} \frac{R^2 \vec{d}_1 \cdot \vec{d}_2 - 3(\vec{R}\cdot\vec{d}_1)(\vec{R}\cdot\vec{d}_2)}{R^5}$$